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FINANCIAL STABILITY

Digital Assets: EU regulatory framework, market uptake, risks and challenges

This briefing provides background on digital assets ahead of the ECON public hearing on 3 December 2025. In addition to providing quantitative and qualitative background information on digital assets including stablecoins, the briefing also aims to clarify how the decentralized and centralized parts of crypto assets interact. It offers two main take-aways: First, since the creation of Bitcoin as the first genuine digital asset, alternatives have in general re-introduced risks by imitating business models of traditional centralized financial institutions. Second, stablecoins, as a manifestation of this trend, combine elements of decentralized and centralized finance. While they use services of decentralized public ledgers (such as Ethereum) they are (highly) centralized when it comes to corporate decisions (such as minting) or the management of reserve assets and resemble Money Market Funds (MMFs) but with increased operational and cyber risks.

1. What are digital assets ?

The EU's main legislation on digital assets, the Markets in Crypto Assets Regulation ([MiCAR](#)), has the name "crypto" in its title. The draft U.S. [legislation](#) has chosen instead the term "digital assets" as in the title of the public hearing.¹

¹ https://www.agriculture.senate.gov/imo/media/doc/bipartisan_market_structure_discussion_draft.pdf

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Definitions

- **Crypto assets** (EU) vs. **digital assets** (US)

Crypto assets: The EU's MiCAR, defines (Article 2) "crypto-asset" as "a digital representation of a value or of a right that is able to be transferred and stored electronically using distributed ledger technology or similar technology". Based on this definition, MiCAR defines **two types of stablecoins**: an "asset-referenced token", or "ART" is defined as "a type of crypto-asset that is not an electronic money token and that purports to maintain a stable value by referencing another value or right or a combination thereof, including one or more official currencies"; the other type of stablecoin is the "electronic money token" or "e-money token" (EMT) which according to MiCAR is "a type of crypto-asset that purports to maintain a stable value by referencing the value of **one** official currency".

Digital assets: the U.S. has a shared definition in GENIUS Act, Section 2 (Definitions), paragraph (6) and proposed general crypto legislation of "digital asset" meaning "any digital representation of value that is recorded on a cryptographically secured distributed ledger."

- **DLT** (EU) vs. **blockchain** (US)

DLT: The EU defines "DLT" (both [DLT Pilot Regime](#) and [MiCAR](#)) where a "distributed ledger technology" or "DLT" means "a technology that enables the operation and use of distributed ledgers"; "distributed ledger" means "an information repository that keeps records of transactions and that is shared across, and synchronised between, a set of DLT network nodes using a consensus mechanism"; and "consensus mechanism" means "the rules and procedures by which an agreement is reached, among DLT network nodes, that a transaction is validated".

Blockchain: Instead of DLT, the proposed U.S. definition (Title I - Definitions; Rulemaking) (our highlighting in bold) defines "**blockchain**" meaning (A) any technology (i) where data is (I) shared across a network to create a distributed ledger of independently verifiable transactions or information among network participants; (II) linked using cryptography to maintain the integrity of the distributed ledger and to execute other functions; and (III) propagated among network participants to reach consensus on the state of the distributed ledger and any other functions; and (ii) **composed of source code that is publicly available**; and (B) any similar technology to a technology described in subparagraph (A)." For the first time there is a clarification of the legal status of the biggest group of crypto assets (93% of the crypto asset market) in the proposed U.S. legislation put forward in the U.S. Senate, and it makes the explicit carve out for the new type of payment stablecoin under the GENIUS Act.

Digital commodity: Proposed U.S. definition (Title I - Definitions; Rulemaking): "digital commodity" means any fungible digital asset that can be exclusively possessed and transferred, person to person, without necessary reliance on an intermediary, and is recorded on a cryptographically secured public distributed ledger; "digital commodity" **does not include** securities, security derivatives, **permitted payment stablecoins under the GENIUS Act**, banking deposits, commodities, commodity derivatives, pooled investment vehicles, goods, collectibles and other noncommodity assets." Hence, **under the proposed U.S.**

legislation bitcoin is a commodity ending legal uncertainty since its creation in 2009; in contrast, under MiCAR the legal status of bitcoin remains uncertain.

Market uptake including statistics

The value of unbacked crypto assets, such as Bitcoin and Ethereum is primarily driven by market demand (Figure 1) and remains highly volatile. Stablecoins, on the other hand, are designed to maintain a stable value to a reference asset, in most cases the USD.

Figure 1. Recent development of market capitalisation of main types of crypto assets



Source: ECB Financial Stability Report based on CoinDesk data, IntoTheBlock and ECB calculations.

Notes: "Stablecoins" covers the combined market capitalisation of 24 US dollar-denominated stablecoins. "Other" covers more than 10,000 crypto-assets tracked across more than 300 crypto-asset trading platforms. (January 2020–May 2025; left-hand scale: USD trillions, right-hand scale: percentages)

Based on a recent [reply](#) by the ECB to MEP De Masi, euro area households currently have limited exposure to crypto-assets. The ECB's November 2024 Consumer Expectations Survey (CES) covering selected euro area countries indicated that, on average, 9.7% of survey respondents or someone in their household owned crypto-assets with country-specific figures ranging between 6% and 21%. Most owners of crypto-assets have a relatively small exposure: of the respondents who reported holdings, 54% reported holdings of below EUR 1,000 and 91% reported holdings of below EUR 20,000. Aggregate holdings are estimated to represent about 0.23% of households' financial assets.

The cryptocurrency market has continued to expand significantly over recent years (Figure 1). As of Q3 2025, total market capitalisation of cryptocurrencies had reached an all-time high of USD 4 trillion, at the time of writing it had dropped to USD 3 trillion. Bitcoin remains the dominant asset at a current market cap of USD 1.7 trillion at the time of writing ([CoinGecko](#), 2025).

The stablecoin segment accounts for less than 7% of the total cryptocurrency market at an estimated EUR 265–280 billion ([CoinGecko](#) 2025). Despite their relatively small share of total market value (see Figure 1), stablecoins play a disproportionate role in market activity, representing approximately 80% of the total trading volume on major centralized crypto exchanges ([Federal Reserve Board](#) 2025) because stablecoins serve as a ‘quote currency’, or the price in which assets are expressed and traded, for many crypto-asset trades.

Approximately 75% of stablecoin market capitalisation and 95% of daily trading volume are concentrated in two dollar-denominated stablecoins: Tether’s USDT and Circle’s USDC ([CoinGecko](#) 2025). Both coins are considered ‘fiat-backed’ meaning they are collateralised by USD reserves, mostly short-term treasuries. The euro-denominated stablecoin market is comparatively small, at EUR 493.7 million (around 0.1% of the stablecoin market). Stablecoin transactions are conducted mainly on the following blockchains: Ethereum (40%), Tron (34%), Binance (10%), Solana (7%) (VISA, 2025). Much like the rest of the market, these proportions vary notably month to month.

Globally, stablecoins now move hundreds of billions of dollars each month, and regulators are tightening rules to keep pace ([IMF](#)). There is some convergence across jurisdictions with regards, to core requirements—full reserve backing, redemption at par, segregated custody, and strict AML/CFT compliance—but rules differ on who can issue stablecoins, minimum capital levels, and currency-pegging rules. Regulatory goals are broadly aligned: maintaining financial stability, protecting consumers, preventing illicit finance, and safeguarding monetary sovereignty. Global bodies reinforce this direction, with the FSB promoting harmonised principles and the [BIS warning](#) that unregulated private stablecoins pose systemic risks.

Cross-border payments and remittances remain the most mature international use case. Global remittance flows amount to about USD 700 billion annually, according to the [World Bank](#) and the [UN](#), and while precise shares are difficult to quantify, so far only a small—but steadily growing—portion currently moves over blockchain-based rails. An [IMF working paper](#) (2025) estimates that global stablecoin flows exceeded USD 2 trillion (EUR 1.84 trillion) in 2024, with a significant share used in cross-border transactions, particularly in Latin America, the Caribbean, Africa, and the Middle East. Research from the [BIS](#) (also [link](#)) and [IMF](#) indicate that stablecoins can reduce frictions in cross-border payments bypassing correspondent-bank chains, enabling faster settlement and lower fees in certain corridors. [IMF geographic analysis](#) further shows that, relative to GDP, stablecoin flows are especially pronounced in Latin America and the Caribbean, Africa, and the Middle East.

NGOs are increasingly turning to stablecoins for deployments into regions plagued by banking disruptions, capital controls, or volatile FX spreads. Yet these tools bring their own challenges, including privacy risks, liquidity constraints, tax uncertainty, and the need for robust compliance by local partner organisations. The [UN](#) (UNHCR, UNICEF and UN World Food Programme) now use blockchain-based

voucher and identity systems for more than one million beneficiaries, generating administrative savings of 1–1.5%.

Apart from U.S. and EU, offshore financial centres continue to support global tokenisation and stablecoin activity. Singapore, Hong Kong, Dubai, and the Cayman Islands offer either explicit regulatory regimes or more flexible licensing requirements. [Dubai's VARA](#) had licensed more than 1,000 digital asset entities by 2024, while [Singapore's MAS](#) has supported more than 20 regulated DLT pilots focused on tokenised bonds, cross-border settlement, and digital asset trading. While these frameworks provide predictability for firms seeking clarity on reserve rules, redemption rights, and disclosure obligations, they also facilitate regulatory arbitrage, where issuance may occur in one jurisdiction while reserves are held in another.

Banking consortia are also launching stablecoin projects: Ten major banks (Bank of America, Deutsche Bank, Goldman Sachs, Citi, UBS, TD Bank, MUFG, Barclays, Santander and BNP Paribas) are jointly exploring issuing a stablecoin pegged to G7 currencies ([link](#)). A consortium of European banks (ING, UniCredit, Banca Sella, KBC, DekaBank, Danske Bank, SEB Caixabank, and Raiffeisen Bank International) is exploring issuing a euro-denominated stablecoin ([link](#)). Swift stands to launch blockchain / create a shared digital ledger to facilitate cryptocurrency transaction including stablecoins ([link](#)). Visa and Mastercard explore ways to introduce stablecoins into payment structures ([link](#), [link](#), [link](#)). Finally, Circle and Deutsche Börse announced partnership to use Circle's stablecoins on Deutsche Börse's financial market infrastructure ([link](#)).

Retail international use of stablecoins is growing where currency instability or FX controls constrain households. In parts of [Latin America](#) and [Africa](#), USD-linked stablecoins serve as informal hedging tools for individuals navigating high and volatile inflation rates often combined with restricted access to foreign currency, i.e., users in such markets hold stablecoins primarily to preserve purchasing power.

2. A brief look at the underlying technology

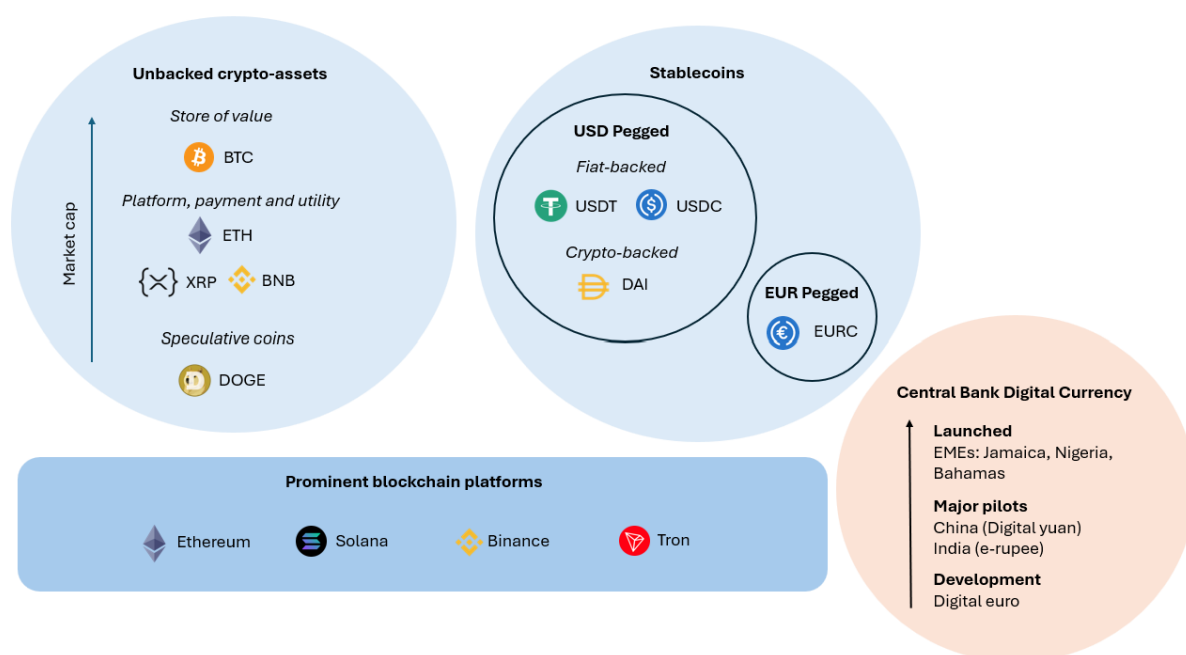
Public blockchains provide transaction validation and time stamping services

The two basic (mathematical) innovations for blockchains' validation and time stamping services provided on public blockchain ledgers date back to work in cryptography in the mid-1980s and early 1990s. Two seminal papers are the 1985 [paper](#) on zero-knowledge proofs by Goldwasser, Micali, and Rackoff ([link](#)), and the 1991 [paper](#) by Haber and Stornetta on cryptographically time stamping a document ([link](#)). Finally, [Bitcoin](#) leveraged the existing physical infrastructure as the internet as a vast system of connected computing resources.

Blockchains are a foundational layer providing transaction validation and time stamping services for crypto assets. In 2015, the Ethereum blockchain launched and became the basis for many DLT-based applications. With the two complementary public ledgers up and running, digital assets now became a real possibility. It was basic science innovations, the self-governing open source model and the innovative economics of Bitcoin that *cumulatively* created an environment where digital assets can thrive. Figure 2 provides an overview: representing blockchains as a foundational layer providing the transaction validation

and time stamping services for **two types of crypto assets** (or digital assets in the U.S. terminology): **unbacked** crypto-assets such as bitcoin, and **backed** crypto-assets such as stablecoins. As can be seen from Figure 2, blockchain platforms such as Bitcoin², Ethereum, Solana, Binance and Tron, serve as infrastructure for the issuance and transfer of unbacked cryptocurrencies, stablecoins, and other digital assets. Finally, Central Bank Digital Currencies (CBDC) are 'digital legal tender' issued on a blockchain by a central bank.

Figure 2. Overview of the crypto-asset ecosystem



Depending how much level of institutional trust can be leveraged in a particular environment, financial use cases have created many different hybrid systems combining centralized and decentralized systems. Such systems preserve certain advantages of public blockchains, e.g., in terms of auditability, increased difficulty for a single party to manipulate the ledger, etc. **Stablecoins are an extreme version of such hybrid systems.** They use public blockchains (such as Ethereum) for decentralized transaction validation and timestamping services but are fully centralized with regards to the minting of the stablecoins themselves as well as the holding of reserves. As we will see in Section 4, the difference between minting and issuance is critical. Minting describes the actual procedure of creating a given amount of stablecoins: this is a transaction registered on the blockchain used by the stablecoin issuer. Issuers do not run their own blockchain but rely on existing public blockchains for those transaction validation and time stamping services.

Much of post-bitcoin innovation looks like a departure from bitcoin's genuine innovations, opting instead for ever higher levels of centralization and concentration of risk. Public and private blockchains

² Bitcoin should be also included in the bottom layer "Prominent blockchain platforms" in Figure 2. Even if it more popular as an asset class, it is the foundational blockchain platform.

give rise to many interesting use cases despite their computational inefficiency. Money counts among them, but bitcoin demonstrates that it is a costly one, and for many uses it cannot compete with fiat money on cost terms alone. Too many crypto finance projects claim to have found computationally less onerous solutions compared to bitcoin only to create the same types of moral hazards that bitcoin's source coders denounced.

Traditional financial institutions' exposure to crypto and increasingly hybrid market designs

The [ECB's Financial Stability Review of May 2025](#) showed a still limited but increasing interconnectedness with traditional finance. By the end of 2024, the direct holdings of euro area significant institutions of crypto-assets stood at around EUR 1 million, up from around EUR 66,000 in 2023. Similarly, exposures to derivatives with crypto-assets as the underlying grew between 2023 and 2024, from around EUR 400 million to EUR 600 million. More significant are the indirect exposures to crypto-assets through services such as custody and deposit-taking from crypto-related businesses. Euro area significant institutions increased tenfold custody services related to crypto-assets and crypto-related investment products in 2024, reaching EUR 4.7 billion up from around EUR 400 million in 2023. Moreover, banks increasingly offer crypto-related services such as brokerage and trading.

While (decentralized) crypto finance was launched as an alternative to traditional finance with its centralized asset pools, recent market players increasingly combine decentralized and centralized designs. They could end up compounding negative risks of both systems. As a telling example, crypto exchanges can be centralised or decentralised. In a centralised exchange the exchange functions as a bank, holding your assets. Out of this list, only HTX (HKEX) and Coinbase (NASDAQ) are publicly traded companies, the majority of exchanges are private. Many of these private firms have highly concentrated ownerships models. None of these recent hybrid solutions has already stood the test of time over a full monetary policy, business, or asset price cycle. **Recent hybrid models have yet to demonstrate their robustness during a severe economic downturn or a major correction in asset markets** as asset markets have been booming since the highly accommodative monetary policies after the the Great Financial Crisis.

3. EU legislative framework

In the EU, crypto-asset markets are now covered by a combination of the new **Anti-Money Laundering Regulation**, AMLR³, the **Sixth Anti-Money Laundering Directive**, AMLD6⁴, the **Transfer of Funds Regulation**, TFR⁵ and the **Markets in Crypto-Assets Regulation**, MiCAR⁶.

³ [Regulation \(EU\) 2024/1624 of the European Parliament and of the Council of 31 May 2024 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing](#)

⁴ [Directive \(EU\) 2024/1640 of the European Parliament and of the Council of 31 May 2024 on the mechanisms to be put in place by Member States for the prevention of the use of the financial system for the purposes of money laundering or terrorist financing, amending Directive \(EU\) 2019/1937, and amending and repealing Directive \(EU\) 2015/849](#)

⁵ [Regulation \(EU\) 2023/1113 of the European Parliament and of the Council of 31 May 2023 on information accompanying transfers of funds and certain crypto-assets and amending Directive \(EU\) 2015/849](#)

⁶ [Regulation \(EU\) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto-assets, and amending Regulations \(EU\) No 1093/2010 and \(EU\) No 1095/2010 and Directives 2013/36/EU and \(EU\) 2019/1937](#)

Table 1: The rapidly evolving U.S. legal framework for crypto assets: Comparing (adopted as well as draft) U.S. legislation – EU legislation on stablecoins, crypto assets in general, and digital (central bank) money

	U.S.	EU	Main differences
Stablecoins (less than 10% of all crypto assets in circulation)	GENIUS Act; at this stage not applicable before 2026Q2 given outstanding required rulemaking to implement the provisions of the Act	MiCAR (issuance of stablecoins), PSD2, EMD, SFD	MiCAR redemption at par at any time and without fees; minimum 30%-60% of reserve assets in the form of bank deposits
Crypto assets other than stablecoins (more than 90% of crypto assets)	Draft U.S. legal framework on crypto assets other than stablecoins authorized under the GENIUS Act with enhanced supervisory competence for CFTC for commodity spot markets; Bitcoin to become a commodity under U.S. law	MiCAR, EU legislation defining supervision of trading venues and trading infrastructure; the legal status of the most important crypto asset Bitcoin remains uncertain in the EU and differs across member states; to be seen how the 3 Dec package on trading and market supervision will impact this situation	Substantially different legal traditions: in the U.S. there are formal tests to decide if a crypto asset is a security (Howey test) or a commodity. MiCAR has a different (functional) approach and cannot be compared directly; not clear what this implies for possible equivalence regimes
Central bank digital currencies (CBDCs)	Draft legal proposal (in the U.S. lower chamber, House of Representatives) on outruling a CBDC	Legal proposal on the establishment of digital euro currently with co-legislators	Total opposition: U.S. planning to outlaw central bank digital money; there is no information in the public domain for the time being how this could impact the international use of the digital euro

The U.S. has adopted a stablecoin specific legislation with the [GENIUS Act](#) on 18 July 2025 including a proposed ban of a U.S. central bank digital currency.⁷ In the meantime, on 10 November, the competent Senate Committee has put forward a [bipartisan draft legal act](#) to regulate the majority of digital assets in the U.S. (i.e., more than 90% of crypto assets that are not falling under the GENIUS Act). In this proposal the create a new competence for the U.S. Commodity Futures Trading Commission (CFTC)⁸ for commodity crypto assets (incl. Bitcoin) while confirming the Securities and Exchange Commission (SEC) as the competent authority for regulating crypto securities. Trading venues, exchanges, etc., all become regulated entities similar to traditional finance market agents and infrastructures.

Table 1 compares the three recent U.S. legal proposals with the current EU legislation. The two pieces of legislation, GENIUS Act and MiCAR, target very different outcomes and, as the [paper](#) by Gabor argues, they convey very different objectives and intentions of the respective legislator. The GENIUS Act establishes⁹ a legal framework for U.S. issued stablecoins for the use in payments and settlements. It is a

⁷ See this [study](#) for a detailed comparison of the U.S. and EU legal frameworks published shortly before the final adoption of the GENIUS Act in the context of the June Monetary Dialogue. The 20 October ESRB report also contains a short comparison (pages 32–33).

⁸ See this 18 November [speech](#) for an overview of most recent action by the CFTC on (de-)regulating digital assets: <https://www.cftc.gov/PressRoom/SpeechesTestimony/opapham19>

⁹ The legislative work triggered by the Genius Act is far from finished: But only after the summer has the U.S. Treasury launched the process of rule making that will operationalize the new regime. An open public consultation on almost all elements in the act, including its core definitions, only ended on 4 November. Other relevant federal rule makers including the Office of the Controller of the Currency (OCC) and the Federal Reserve System (the U.S. Fed in its function as the federal banking supervisor) have not even launched this necessary rulemaking process. The

legal framework for privately issued crypto money, but allowing more freedom as to the composition of reserve assets. Also, the GENIUS Act has explicitly extra-territorial reach and is binding on any stablecoin issuer offering stablecoins to a U.S. person. Despite these differences, the creation of a possible equivalence regime with the U.S. was already indicated by the ECB in its April non-paper as a possible solution.

A comparison of the two legal frameworks shows: issuance of third country MISCs tends to be more profitable in the U.S. because there is currently¹⁰ no minimum requirement to hold at least 30% of reserves in bank deposits as under MiCAR, **and this requirement increases to 60% in the case of a significant issuer**;¹¹ yields on reserve assets are higher in the U.S., and the trade-related demand for the coin is higher as well. In addition, U.S. based stablecoin issuers continue to operate in a partly unregulated environment and benefit from considerable regulatory forbearance in the current setting. Even after U.S.-based issuers will become regulated under the GENIUS Act, they will not be subject to the more onerous composition of reserve assets requirements that exist under MiCAR.

4. Stablecoins: how risky are they?

Comparison of the business models of money market funds (MMFs) and stablecoin issuers

Both MMFs and stablecoin issuers engage in various forms of liquidity and maturity transformation in order to make money. The [ECB's latest Financial Stability Report](#) also mentions similarities in terms of their vulnerabilities to market risk. However, in the details there are many differences, including concerning legal form, liability structure, sourcing of liquidity, regulatory treatment and, of course, operational mechanics. The question is how those differences translate into different risk profiles, or if, to the contrary, despite numerous differences both business models generate similar risk profiles and share problematic economic incentive structures. Below we have compiled a list of key differences.

Regulation: MMFs are regulated in the EU under the [MMF regulation](#). In contrast, stablecoins that do not qualify as securities are regulated under the [Markets in Crypto Assets Regulation](#) (MiCAR).

Value proposition to investors/users: MMFs offer market-based liquidity and still pay a yield, i.e., the shareholders share the returns from liquidity and maturity transformation with the operators or arrangers of the fund. They offer a flexible solution for institutional and retail investors alike that need to go into cash while waiting for the next trading opportunity. MMFs still offer a small yield for those holding periods. In contrast, stablecoins have a different value proposition and as a rule they do not pay yields (see also the section on comparing U.S. GENIUS Act and EU MiCAR). The user only gets the “benefit” of a 1:1 peg between

U.S. Genius Act will enter into application at the earliest 120 days after this rule making has been finalized. Stablecoin issuers would then be able to issue Genius Act compliant coins as soon as they have received authorisation after that period. Authorisation can be granted at federal or state level. The Genius Act also establishes a new federal governance framework with the U.S. Treasury acting as the chair of the new federal body.

¹⁰ Currently, U.S. issued stablecoins are not regulated with regards to composition of their assets, and even after the GENIUS Act will apply there will not be a minimum requirement to hold a certain share of reserves in the form of bank deposits; this is not saying that currently existing stablecoin schemes do not use bank deposits.

¹¹ These 30% (non-significant issuer), 60% (significant issuer) requirements apply to both types of stablecoins under MiCAR (ARTs, and EMTs).

a crypto asset and a traditional fiat money asset (bank deposits that may or not benefit from a government protection, or short term government debt instruments). The stablecoin issuer keeps all of the economic rent from liquidity and maturity transformation. Stablecoins do not look like a very attractive value proposition until one takes into account how cumbersome the transition from crypto assets to traditional (fiat money) assets remains. The on-ramp, off-ramp function of stablecoin therefore is an attractive value proposition for all those seeking exposure to more risky crypto assets. As the ECB [noted](#) in its latest Financial Stability Review, **stablecoins are now part of 80% of trades executed on crypto trading platforms**. As long as the issuer can ensure redemption at par (or at least as long as the market believes that the issuer can ensure redemption at par) the stablecoin issuer collects a digital seignorage by minting digital money (how much the issuer can mint is precisely one of the un-resolved operational and regulatory issues). In addition to returns on reserve assets, the stablecoin issuer may generate other incomes in exchange for on-chain services, arbitrage or market-making profits, or revenues from commercial cooperations.

How to get the money out: MMFs use traditional banking and payments rails, off-balance sheets assets are held by custodians; all these roles are highly regulated in the U.S., the EU, and other financially developed jurisdictions. Stablecoin issuers have a more complex setting due to their function of providing a bridge between crypto asset environment on-chain and fiat asset environment off-chain. **Stablecoin reserve assets are off-chain, digital tokens are on-chain**. They are not directly linked (unit per unit) as in the case of tokenized off-chain assets (e.g., tokenized deposits). Having only the corporate balance sheet and the promise of redemption at par as a link the stablecoin issuer keeps the issuer very flexible in terms of daily operations. The flexibility is also linked to weaker oversight and regulation as the next point discusses.

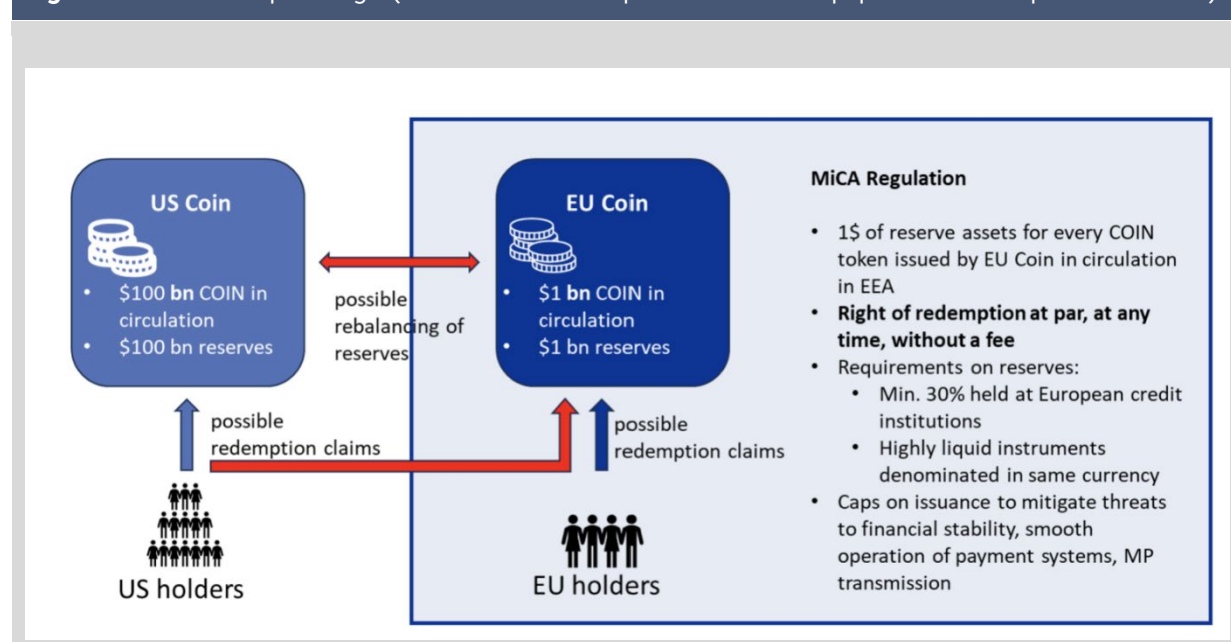
Transparency, controls, audit: MMFs are regulated entities with clearly established rules for audit, disclosure, and reporting. Stablecoin issuers have high levels of transparency on-chain where the use of public blockchains ensures full visibility of on-chain transactions; however, the off-chain reserve assets are not subject to harmonized rules and it will depend on the initiative of the issuer how often and how detailed the composition of reserves and custodial arrangements of such reserves are disclosed and reported, or audited. Circle, the second largest stablecoin issuer, became a publicly listed company in the U.S. in 2025. As a result the publish quarterly financial reports.

Summing up: Both models earn a difference between reserve yields and what shareholders/users receive (with stablecoin users typically not receiving anything); but MMFs are more constrained as concerns the division of returns and face higher levels of investor protection. Stablecoin issuers are more free to generate revenues from different on-chain and off-chain sources, and higher operational risks seem to fall more on the side of users that are less protected due to a lack or uneven regulation.

Already in April, the ECB had alerted Member States about the risk of third country multi-issuance stablecoins (MISCs) in the form of a [non-paper](#) tabled to the Council's Financial Stability Committee (FSC) of 14 April. In the meantime, the Commission and the ECB tabled another set of non-papers to the [Eurogroup meeting in Luxembourg on 9 October](#); however, and the topic was again on the [agenda](#) of the Eurogroup on 12 November, when Eurogroup president Donohoe [announced](#) the creation of a dedicated

work stream on stablecoins after the meeting. Also, the recent INI [report](#) on financial stability voted in the last ECON Committee on 5–6 November¹² raises similar financial stability concerns.

Figure 3. MISC redemption logic (ESRB 20 October report and ECB non-paper for the 14 April Council FSC)



Notes: The diagram conceptualises the flow of reserves and redemption claims between an EU and a US stablecoin issuer. It highlights how tokens can move and be redeemed in either jurisdiction, while underlying reserves remain subject to two regulatory and supervisory silos, increasing the risk of misalignment in crises.

Source: ESRB, Report on stablecoins, crypto-investment products and multi-function groups, October 2025.

On 27 October, Richard Portes published an [extended version](#) of his previous FT article on the regulatory arbitrage of third country multi-issuance stablecoins (MISCs) under MiCAR. In an external [paper](#) requested by the ECON Committee, Daniela Gabor arrives at similar conclusions: the current U.S. – EU regulatory approaches create a situation where **MISCs live in the more profitable jurisdiction (the U.S.), while any redemption runs would first attempt to deplete reserves in the EU where redemption rights are formulated in a stronger way** (right of redemption at par at any time, without a fee). As both Portes and Gabor explain, the **situation of regulatory arbitrage** arises in the following way: a MISC issues a coin in the U.S. and a coin in the EU. They are fully fungible and (technically and legally) indistinguishable for the user (the bank or the payment service provider for that matter). As the coin is pegged to the U.S. dollar, the corresponding coin issued in the EU is a digital token form of what is typically called a *eurodollar* in fiat money markets. One can speak of *eurodollar stablecoins*.

Figure 3 depicts in a stylized manner the situation between the two jurisdictions: the U.S. has higher issuance volumes than EU. During normal times, this does not necessarily create issues and redemptions take place in both jurisdictions. Issuance is more profitable in the U.S. because there is no minimum requirement to hold at least 30% of reserves in bank deposits as under MiCAR. Bank reserves of at least 30% (or 60% for significant issuers) under typical market conditions work like a cap on profits for stablecoin issuers. Instead,

¹² Safeguarding and promoting financial stability amid economic uncertainties, 2025/2051(INI), Rapporteur: Johan Van Overtveldt

the U.S. issuer can invest in a wide range of high quality liquid reserve assets; the GENIUS Act will preserve this advantage for U.S. issuers as it allow not only short term U.S. Treasury bills, but also long term government bonds (that pay higher yields) as long as their remaining maturity is 93 days or less. Hence, the whole range of government debt securities is in scope that will often pay yields significantly above interest on bank deposits.

However, **during market stress the issuer may ring fence its reserves in the U.S.** thus creating a possible mis-match between the availability of reserves and the desire to redeem digital tokens in the EU (with its stronger protection of redemption rights under MiCAR). In particular, Portes recalls that such ring fencing took place also within the EU during liquidity crunches both after the Lehman failure in autumn 2008 and during the pandemic in spring 2020. It is therefore prudent to calculate risk scenarios where ringfencing will take place on either side. Portes and Gabor continue to outline the **macroprudential risks implied by such a redemption run scenario**: Some EU banks may have difficulties to ensure redemption at par, without delays, and without cost for the user asking for it. The emergence of banks specializing in crypto finance could lead to a small group of banks with high exposures. It then suffices that a few of those banks have concentrated funding links with other, more traditional banks for contagion to spill over into the traditional banking sector.

There are two elements in the ESRB [Recommendations](#) that directly concern MiCAR and the co-legislators (while other elements need to be addressed by competent authorities): the request to the Commission to provide a binding interpretation of Union law as concerns MISCs, and the (possibly accelerated) timing of the general evaluation MiCAR possibly accompanied by a legislative proposal. Specifically, the ESRB asks the Commission to either clarify (by end of 2025) if this type of MISC is permitted under MiCAR, or bring forward the report to Union legislators due by June 2027 (Art 140, MiCAR). The ESRB's recommendations are embedded in a rich set of analyses by experts in relevant international bodies¹³ and academics¹⁴.

Stablecoins and the transmission of monetary policy

In a recent CEPR study, [Aldasaro et al. \(2024\)](#) compared how stablecoins behaved under stress compared to money market funds (MMFs). MMFs have encountered their own financial stability issues that are reviewed in a separate [EGOV briefing](#). In that study, the authors note that **stablecoins at first sight look like the digital asset version of MMFs**; like MMFs they create profit via liquidity and maturity transformation (see Annex I for an overview of main differences). **However, MMFs and stablecoins behave very differently under stress.** Most importantly, stablecoins amplify monetary policy shocks and thereby run counter the objectives of the monetary policy maker. Specifically, a tightening of monetary policy (in this case an increase in the main policy rate) leads to a negative reaction by stablecoins, whereas the MMF gains in value. The MMF, where shareholders participate in the profits generated by liquidity and maturity transformation,

¹³ See, e.g., chapter 3 of the recent [BIS Annual Report](#)

¹⁴ See e.g. <https://cepr.org/voxeu/columns/multi-issuer-stablecoins-threat-financial-stability>, <https://cepr.org/voxeu/columns/stablecoins-and-money-market-funds-less-similar-you-think>, <https://www.imf.org/en/publications/fandd/issues/2025/09/stablecoins-tokens-global-dominance-helene-rey>; https://www.europarl.europa.eu/cmsdata/296400/DEZERNAT%20June%202025_FINAL.pdf ; https://www.europarl.europa.eu/cmsdata/296480/MD_SFL%20June%202025_FINAL.pdf ; <https://www.project-syndicate.org/commentary/us-stablecoin-law-genius-act-risk-to-financial-stability-democracy-sovereignty-by-hilary-j-allen-2025-09>

is more robust after the increase in the policy rate as the tightening of policy benefits the liquidity of high quality short term assets (substitution effect) and increases the return on such assets (price effect). Stablecoins instead come under pressure which discloses that market participants use them differently, including for leveraged (crypto) trades that become less profitable as the policy makers increases the price of credit at the short end of the yield curve.

Recent months have also seen growing debate over how an expanding stablecoin market could affect the euro's monetary sovereignty. This issue was also explored in expert analyses commissioned by the ECON coordinators for the European Parliament's June Monetary Dialogue. The [Monetary Policy Expert Panel's](#) contributions provide complementary yet sometimes contrasting views on the influence of dollar-pegged stablecoins and the wider consequences of the rise of "on-chain money." All three papers—published on the [MD website](#)—recognise the systemic importance of dollar-denominated stablecoins and broadly describe the U.S. regulatory stance as permissive and strategically aimed at strengthening the dollar's global position. Their main differences lie in how they evaluate the scale of the risks involved and in the policy responses they recommend.

[Bezemer et al. \(Sustainable Finance Lab\)](#) adopt a relatively reassuring view, arguing that the adoption of dollar-denominated stablecoins in the euro area is likely to remain limited. They point to the high level of public trust in the euro, the robustness of European payment infrastructures, and the restrictive nature of the EU's [MiCA regulation](#) as effective barriers to large-scale uptake. While they acknowledge the need for continuous monitoring, they see little immediate threat to the ECB's monetary policy framework.

In contrast, [Angeloni and Tille \(CEPS\)](#) seems to take a more cautious view. Although they concede that the euro area is not particularly exposed to dollarisation risks, they argue that unregulated use of stablecoins could nonetheless weaken the effectiveness of monetary policy transmission. They are sceptical of a retail-focused digital euro and instead advocate for a wholesale version designed to enhance cross-border settlement efficiency. In their view, the development of dollar-aligned stablecoins outside of central bank control represents a high-risk strategy that could either modernise global finance or introduce new forms of systemic instability.

[Van 't Klooster et al. \(Dezernat Zukunft – EMPN\)](#) take a more geopolitical and strategic perspective. Authors emphasise the divergence between the US and the euro area in regulating on-chain money, framing the former approach as a form of "cryptomercantilism" aimed at expanding the dollar's international use and increasing demand for US debt. By contrast, they argue that the euro area has prioritised monetary sovereignty and financial stability through the [MiCA regulation](#) and the promotion of a digital euro as a public alternative. They warn that widespread use of dollar-denominated stablecoins – especially in emerging markets – could further marginalise the euro and call for a multilateral response based on CBDCs and fast payment systems. For these authors, euro area must resist calls for deregulation and avoid regulatory equivalence with weaker regimes like the US Genius Act, instead asserting its own values through a cooperative, rules-based infrastructure for digital payments.

More recently, some policymakers, such as Dutch central bank governor and ECB Governing Council member [have warned](#) that if stablecoins turn out not to be truly stable, rapid liquidation of their underlying assets could be required. In such a scenario, the ECB might be forced to adjust its monetary policy stance if stablecoin instability disrupts key economic and financial linkages. This could have adverse consequences not only for financial stability but also for the broader economy and inflation. In these circumstances, a central bank like the ECB could be compelled to respond, potentially through policy tightening or easing, but the direction and magnitude of the response would be highly uncertain, as central banks have not yet faced shocks of this kind.

The same concerns about stablecoins have also surfaced in letters to the ECB. An example is available [here](#) where an ECON committee member asked the ECB a direct question the effect of stablecoins on central bank functions, namely would the ECB be willing to act as a lender of last resort if stablecoin issuers located outside the euro area were unable to meet their redemption commitments—for instance, U.S. providers operating within the shadow-banking system? The letter further mentioned whether the ECB might intervene in such a case in a manner similar to its past support for money market funds facing stress. [In its response](#), the ECB underlined that **the crisis-management framework established under MiCAR is fundamentally different from lender-of-last-resort arrangements**. Specifically, that MiCAR sets out procedures for addressing failures of crypto-asset service providers, whereas lender-of-last-resort functions involve central banks providing liquidity to safeguard the smooth functioning of financial markets and to preserve overall financial stability.

Stablecoins: compounding negative risks of centralized and decentralized finance ?

The fact that private issuers of digital currency do not own or control the underlying “plumbing” of the crypto financial system has a clear advantage: transparency about what has been recorded. On 14 October, the stablecoin issuer Paxos accidentally “minted” no less than USD 300 trillion of US dollar based stablecoins (before “destroying” them about 20 minutes later), as CNBC [reported](#).¹⁵ Because the issuer, an authorized issuer in the EU (Table 2), did not control the underlying infrastructure (the Ethereum blockchain), the 14 October incident was visible to everybody. It confirmed that, as in the case of money market funds (MMFs), there was no direct physical link between the minted coins and underlying reserves. The only links are the balance sheet of the issuer and the promise of redemption at par which, of course, depends entirely on the availability of sufficient reserves (a permanent requirement under MiCAR). But as Lorena Nessi [observed](#): “(...) Even though the tokens were burned within minutes, the scale of the error exposed how concentrated power in centralized systems can turn into systemic risk.”

¹⁵ <https://www.bitget.com/news/detail/12560605017587>

Table 2. MiCAR authorised stablecoins (EMTs) as of 15 September

Stablecoin Name	Issuer	Market Capitalisation as of 15 September 2025 (in USD)	Variation since 10 June 2025 (over three months)	Comment
USD pegged stablecoins				
USDC	Circle	\$73.1 billion	+36%	multi-issuance with U.S. Circle LLC, figure reflects worldwide issuance
USDG	Paxos Issuance Europe	\$592 million	n/a (new EMT)	multi-issuance with Paxos Singapore, figure reflects worldwide issuance
USDCV	Société Générale – Forge	\$32 million	n/a (new EMT)	
USDQ	Quantoz Payments	\$6.6 million	n/a (new EMT)	
USDR	StabIR Ltd	\$5.2 million	-13%	
USDSM	Stable Mint	\$5 million	n/a (new EMT)	
eUSD	Paxos Issuance Europe			being withdrawn after Paxos acquired Membrane, focusing on USDG.
USDE	Eurodollar			no data available
EUR pegged stablecoins				
EURC	Circle	\$238 million	+20%	
EURCV	Société Générale – Forge	\$66 million	+37%	
EURI	Banking Circle	\$56 million	+22%	
EURAU	AllUnity	\$21 million	n/a (New EMT)	
EURR	StabIR Ltd	\$13 million	+8%	
EURQ	Quantoz Payments	\$5.5 million	+44%	
EURØP	Schuman Financial	\$4.1 million	+341%	
EURD	Quantoz Payments	\$949 thousand	+2%	
EUROe	Paxos Issuance Europe	\$183 thousand		being withdrawn after Paxos acquired Membrane, focusing on USDG.
EURSM	Stable Mint			no data available
ENEUR	Fiat Republic			no data available
Ambr EMT	Ambr Payments			no data available
BLUEUR	Blue EMI LT			no data available
GBP stablecoin				
ENGBP	Fiat Republic			no data available
CZK stablecoin				
CZKI	Payment Corporation SE			no data available

Source: ESRB report 20 October, Table 1, page 17

Notes : based on ESMA MiCAR register as of 15 September 2025

Market cap data from CoinGecko, CoinMarketCap, Coinbase, blockchain explorers.

It is critical to keep in mind that **the issuer is a regulated issuer, authorized also in the EU under MiCAR, and there was no issue in terms of know your customer or anti-money laundering measures..** Rather **the incident is a reminder how centralized the minting process itself is**, and there are indeed few other

examples where operational risks are centralized more heavily. The incident should serve as a reminder and a **warning that (legal) compliance alone does not guarantee financial stability or contain cyber security risks** in a satisfactory manner. The blockchain's transparency allowed the error to be caught quickly; at the same time it revealed how under currently used operational protocols easily such events can occur.

The legal fiction of MiCAR is that a stablecoin issuer invests the funds received and then ensures that reserves *at all times* remain at least as high as outstanding stablecoin value. However, MiCAR does not say *when exactly* the stablecoin is minted. Based on reflections after the Paxos incident, there seems to be a good case for requiring that only after proof of sufficient reserves a corresponding amount, and no more, of stablecoins can be minted. But **even if the proof of reserves is systematically included in the minting process, it would only proof that at that point in time sufficient reserves were available**. A short moment later this could no longer be the case.

In the ESRB [report](#) on systemic risks from crypto-assets that assembles the analysis underlying the recommendations on stablecoins, the ESRB refers back to the BIS [2025 Annual Report](#) of 29 June where the **BIS concludes that stablecoins lack several important properties that are critical in the context of payments** (see pages 85ff of the [report](#)). The BIS report says also without any ambiguity that stablecoins definitively allow to send money to someone whose identity has not been identified (see next section on money laundering). The BIS also reminds that stablecoins' borderless nature risks a "race to the weakest regulatory links", and the case of third country multi-issuance is one current example for such a race.

5. AML issues

The EU has moved in recent years from a largely fragmented approach to a [more integrated framework](#) for addressing money laundering risks linked to digital assets. Crypto-asset markets are now covered by a combination of the new **Anti-Money Laundering Regulation** (AMLR¹⁶), the **Sixth Anti-Money Laundering Directive** (AMLD6¹⁷), the **Transfer of Funds Regulation** (TFR¹⁸), and the **Markets in Crypto-Assets Regulation** (MiCAR¹⁹), complemented by the establishment of the **new EU Anti-Money Laundering Authority** (AMLA²⁰). However, despite this more coherent legislative architecture, AMLA will initially oversee only a limited number of high-risk cross-border entities, while most supervisory responsibilities will continue to lie with national authorities, whose resources, priorities and enforcement cultures differ significantly. Moreover, key operational elements will depend on forthcoming secondary legislation and on the development of technological capabilities necessary to monitor decentralised environments, meaning that the **effectiveness of the new framework will only materialise gradually**. Together, these instruments

¹⁶ [Regulation \(EU\) 2024/1624 of the European Parliament and of the Council of 31 May 2024 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing](#)

¹⁷ [Directive \(EU\) 2024/1640 of the European Parliament and of the Council of 31 May 2024 on the mechanisms to be put in place by Member States for the prevention of the use of the financial system for the purposes of money laundering or terrorist financing, amending Directive \(EU\) 2019/1937, and amending and repealing Directive \(EU\) 2015/849](#)

¹⁸ [Regulation \(EU\) 2023/1113 of the European Parliament and of the Council of 31 May 2023 on information accompanying transfers of funds and certain crypto-assets and amending Directive \(EU\) 2015/849](#)

¹⁹ [Regulation \(EU\) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto-assets, and amending Regulations \(EU\) No 1093/2010 and \(EU\) No 1095/2010 and Directives 2013/36/EU and \(EU\) 2019/1937](#)

²⁰ [Regulation \(EU\) 2024/1620 of the European Parliament and of the Council of 31 May 2024 establishing the Authority for Anti-Money Laundering and Countering the Financing of Terrorism and amending Regulations \(EU\) No 1093/2010, \(EU\) No 1094/2010 and \(EU\) No 1095/2010](#)

aim to **ensure that crypto-related activities are subject to broadly the same integrity and transparency standards as traditional finance**, but important questions persist regarding consistency of supervision and the capacity to address decentralised or cross-border structures effectively.

A central step has been the **inclusion of crypto-asset service providers (CASPs) as obliged entities under the new AMLR**. The AMLR expressly extends its scope to CASPs that are already brought into the AML perimeter via MiCAR and the TFR, in order to *“mitigate any risk of misuse of crypto-assets for money laundering or terrorist financing purposes”* (Recital 14 of AMLR). This means that **crypto-exchanges and custodial wallet providers have to perform customer due diligence, monitor transactions and report suspicious activities** on a risk-sensitive basis, in line with FATF [standards for virtual assets and virtual asset service providers](#). In parallel, the TFR introduces a full **“travel rule” for crypto-asset transfers**, requiring information on originators and beneficiaries to accompany transfers whenever at least one payment service provider or CASP involved in the transaction is established in the EU.

The EU has also framed MiCAR as part of a broader effort to align with international standards ([Recommendation 16 on Payment Transparency](#), FATF). **MiCAR fills a regulatory gap by creating an authorisation and conduct framework for a wide range of crypto-assets and CASPs**, but acknowledges that, in the absence of a Union-wide regime, some Member States had already adopted diverging rules for assets falling outside existing financial-services legislation, with risks of fragmentation and regulatory arbitrage (Recital 4 of MiCAR). Former **Commissioner Mairead McGuinness underlined in public statement that MiCA is intended both to protect consumers and to embed safeguards against money laundering and terrorist financing**, and expressed the hope that EU rules *“could become a model for other countries”*.²¹ However, important scope and perimeter questions remain. **MiCAR does not comprehensively cover all forms of digital assets**: certain non-fungible tokens, governance tokens and genuinely decentralised arrangements may fall outside its categories or be subject to lighter regimes, leaving a residual *“no-treatment”* area where AML obligations depend mainly on national transposition and general AML rules. This raises the possibility that sophisticated actors could gravitate towards less regulated structures or jurisdictions, despite the EU’s attempt to create a level playing field.

Self-hosted (or “unhosted”) wallets illustrate another structural tension. The TFR explicitly addresses transfers to and from self-hosted addresses when a CASP participates in the transaction. It defines self-hosted addresses and requires CASPs to collect information on both originator and beneficiary, to assess the money-laundering risk of transfers involving such wallets, and, for higher-risk or higher-value transfers (typically above EUR 1 000), to apply mitigating measures such as verifying ownership of the self-hosted address or obtaining additional information on the source and destination of funds (Article 20 of TFR). The European Banking Authority (EBA) has complemented this with 2024 [travel-rule guidelines](#) specifying what information must accompany transfers of funds and crypto-assets and how payment providers and CASPs should react to missing or incomplete data. Nevertheless, **peer-to-peer transfers between private wallets with no CASP in the chain remain largely outside the effective reach of AML rules**, and the proportionality of imposing stronger controls on self-hosted wallets is [contested in parts of the industry](#). This leaves open the policy question of whether current measures are sufficient to prevent high-risk activity from migrating into fully non-custodial environments.

²¹ See [EP Verbatim report of proceedings of 19 April 2023](#), Strasbourg

Decentralised finance (DeFi) raises similar issues. [FATF guidance](#) stresses that, even in decentralised arrangements, there is often a natural or legal person who can be considered the “owner/operator” and should be treated as a Virtual Asset Service Providers (VASP) for AML purposes (Part Three, para.129). In practice, however, **identifying such an accountable entity can be difficult when protocols are governed by dispersed token-holder communities or open-source developers.** While EU law brings DeFi under the AML perimeter where there is an identifiable CASP or service provider, it does not yet provide a fully operational model for genuinely decentralised structures. This gap is increasingly relevant as the [ESRB](#) has warned²² that financial stability risks from crypto-assets and DeFi are “mounting” as these markets become more integrated with traditional finance.

Supervisory capacity will therefore be critical. The new AMLA is expected to directly supervise a limited number of high-risk, cross-border entities, including selected CASPs, and to drive convergence in supervisory practices, including through the use of advanced technology. At the same time, market-level warnings by European agencies have highlighted the need for vigilance. **ESMA²³ has cautioned that the interconnectedness of crypto markets could have wider repercussions, particularly in stressed conditions.** The ESRB²⁴ and the ECB²⁵ have raised specific concerns about stablecoins, urging closer oversight of “multi-issuer” and foreign-issued stablecoin arrangements and calling for safeguards and equivalence regimes to prevent regulatory arbitrage and runs on EU-based reserves.

Overall, the EU has positioned itself at the forefront of global attempts to integrate digital assets into a robust AML/CFT framework. CASPs are now firmly within the scope of EU AML rules, travel-rule obligations have been extended to crypto-asset transfers, and MiCA provides a structured regime for many tokens and stablecoins.

Nevertheless, **several critical questions remain unresolved: how to handle assets and structures that fall outside MiCA’s perimeter; how far to go in regulating self-hosted wallets and peer-to-peer transfers; how to attribute responsibility within DeFi; and how to manage the interaction between EU-regulated markets and large third-country providers.**

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²² See [Crypto-assets and decentralised finance](#), ESRB, 2025

²³ See [Crypto assets: Market structures and EU relevance](#), p.5, ESMA, 2024

²⁴ See [Crypto-assets and decentralised finance](#), Annex 1, ESRB, 2025; [Recommendation ESRB/2025/9 on third-country multi-issuer stablecoin schemes](#)

²⁵ See speech by Christine Lagarde Speech [Cutting through the noise: exercising good judgment in a world of change](#), ECB, 2025